

Ihsan Ul Haq

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Research Interests

Renewable Energy Integration • Microgrids • Smart Grids • Power Electronics based Power Systems • Intelligent Autonomous Systems • Deep Learning • EV-grid Integration

Education

MSc Energy Engineering (Distinction)

University of Hull, United Kingdom | 2021–2022 | Percentage: 72/100

Dissertation: *A Comparative Analysis of Deep Learning Models for Short-Term Load Forecasting.*

BSc Electrical Engineering

University of Engineering and Technology, Peshawar, Pakistan | 2011–2015 | CGPA: 3.47/4.00

Final Year Project: *Design and Analysis of High Gain Microstrip Patch Antenna Using CST.*

Professional Experience

Co-supervisor (In Collaboration, Associated with UET Peshawar)

AI applications in electrical engineering and power systems | 2023– Present

Primary School Teacher

Government Primary School (GPS) Malakana, Swat | 2020–2021

Subject Specialist (Mathematics)

Garland Model School and College | 2016–2020

Research Intern

PTCL Pakistan | Jul–Aug 2014

Selected Research Projects, Associated with UET Peshawar ([Full Project list →](#))

Title: Machine Learning-Based Renewable Energy Forecasting and Priority Load Management for Smart Energy Systems. Co-supervisor | 2025–2026

- Developed a machine-learning-based framework for renewable-energy forecasting and priority-based load management in smart energy systems.

Title: Sustainable Energy Management in Smart Grids Using Machine Learning. Co-supervisor | 2024–2025

- Collaborated on an ML-based approach for sustainable energy management and intelligent decision-making in smart-grid environments.

Title: Machine Learning-Based Fault Detection in Three-Phase Transmission Lines and Electric Machines. Co-supervisor | 2023–2024

- Collaborated on ML-based fault detection and classification for three-phase transmission systems and electrical machines.

Selected Publications ([Full Publication List →](#))

Journal Articles

Khan, T., Mansoor, M., Iltaf, M., Alam, S., Farooq, M., **Ul Haq, I.**, & Ullah, K. (2026). Machine Learning-Based Renewable Energy Forecasting and Priority Load Management for Smart Energy Systems. *International Journal of Innovations in Science & Technology*, 8(2), 786–802. <https://doi.org/10.33411/IJIST/1881>

Khan, A.R., **Haq, I.U.**, Farooq, M., Rehman, B.U., Ullah, K., Amir, M., Khan, M.K. and Khan, M.I. (2025) 'Sustainable Energy Management in Smart Grids Using Machine Learning', *Spectrum of Engineering Sciences*, 3(7), pp. 1332–1342. <https://doi.org/10.5281/zenodo.16671247>

Qasim, M., Akbarzai, J., Khan, A. R., Majeed, M. E., Farooq, M., **Haq, I. U.**, Rehman, B. U., Ullah, K., & Khan, M. K., "Machine Learning-Based Fault Detection in Three Phase Transmission Lines and Electric Machines", *Zenodo, Jul. 2024. doi: 10.5281/zenodo.16146039*.

Conference Paper

Haq, I.U., Khalid, M. and Manzoor, U. (2023) 'A Comparative Analysis of Deep Learning Models for Short-Term Load Forecasting', *IEEE CASE 2023*, Auckland, New Zealand, pp. 1–7. [doi:10.1109/CASE56687.2023.10260626](https://doi.org/10.1109/CASE56687.2023.10260626).

Technical Skills

Programming: Python, MATLAB, C++

Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, Deep Learning, Computer Vision

Data Science: Data Analysis, Data Visualization

Engineering Tools: CST Studio Suite, EndNote, Microsoft Visio, PowerPoint, PSpice

Selected Certifications

- Renewable Energy Technology Fundamentals — University of Colorado Boulder
- Renewable Power & Electricity Systems — University of Colorado Boulder
- Advanced Learning Algorithms — DeepLearning.AI
- Reinforcement Learning — University of Alberta / DeepLearning.AI
- Mathematics for Machine Learning: Linear Algebra — Imperial College London
- An Intuitive Introduction to Probability — University of Zurich
- Python Data Structures — University of Michigan
- Convolutional Neural Networks — DeepLearning.AI

Awards & Memberships

- NICT Merit Scholarship, UET Peshawar (2011–2015)
- Pakistan Engineering Council (PEC), Membership No. 48533
- Saudi Council of Engineers, Membership No. 305735
- Academic Excellence Award (2010)

Languages

Pashto (Native) | English (Advanced) | Urdu (Advanced)

References

References available upon request